



Quantifying allosteric stabilization in protein-protein interactions with enhanced sampling

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Undergraduate Research Opportunities

Max Planck School Matter to Life

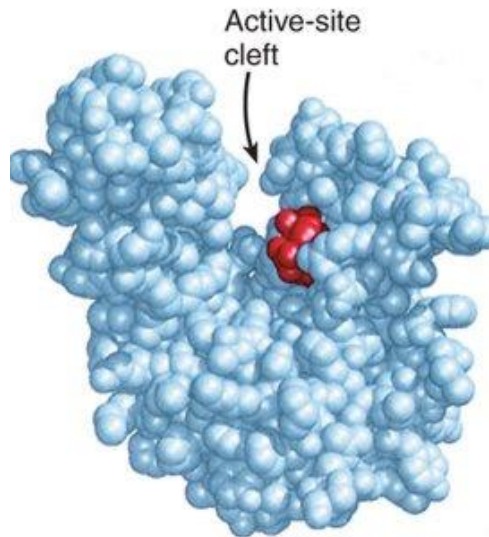
Technische Universität München

The Hong Kong University of Science and Technology

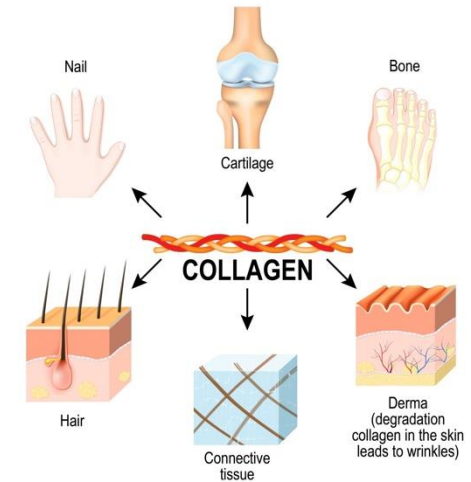
Proteins are the workhorse of our body



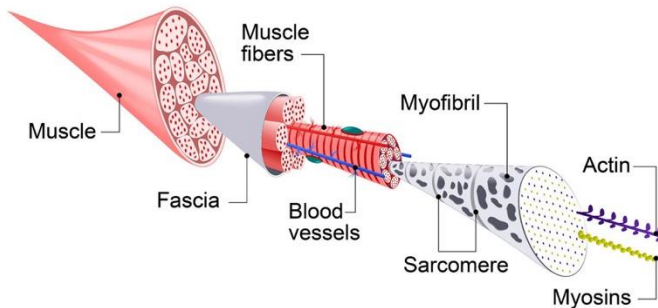
Hemoglobin in red blood cells **transport** oxygen



Enzymes are large proteins that **catalyze** reactions



Collagen is a protein that gives **structure**

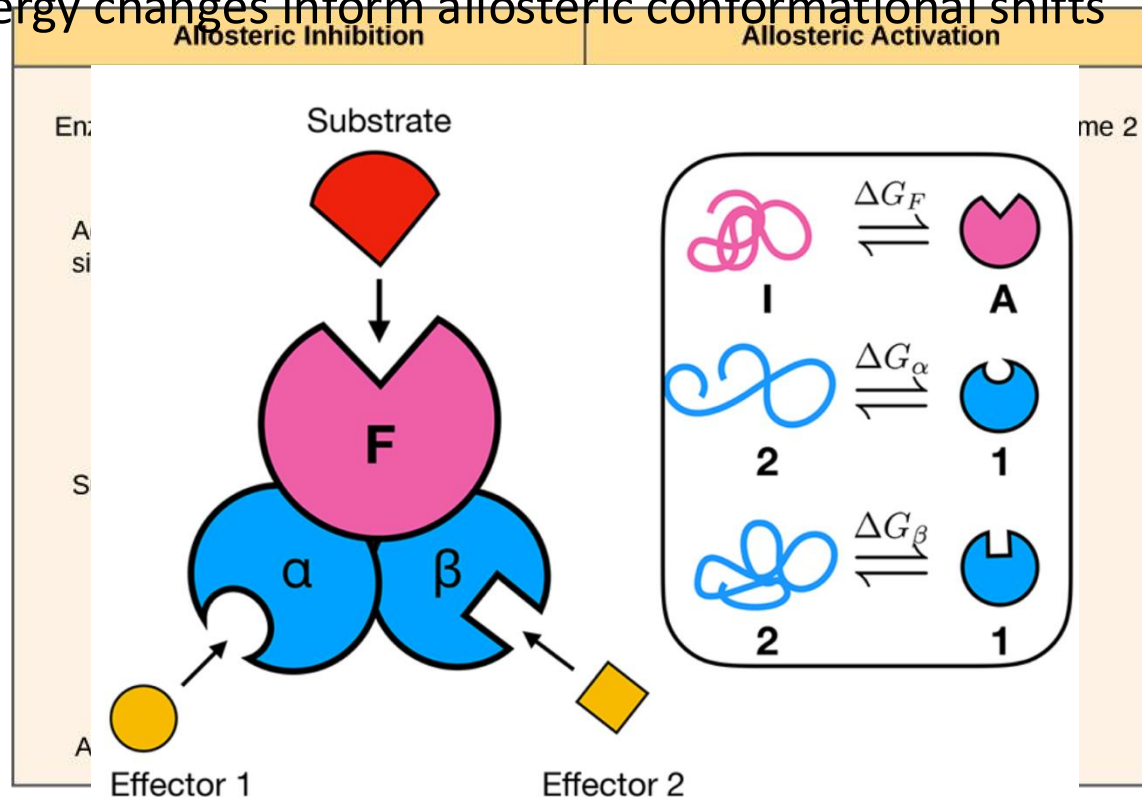


Muscles are made of proteins that gives **movement**

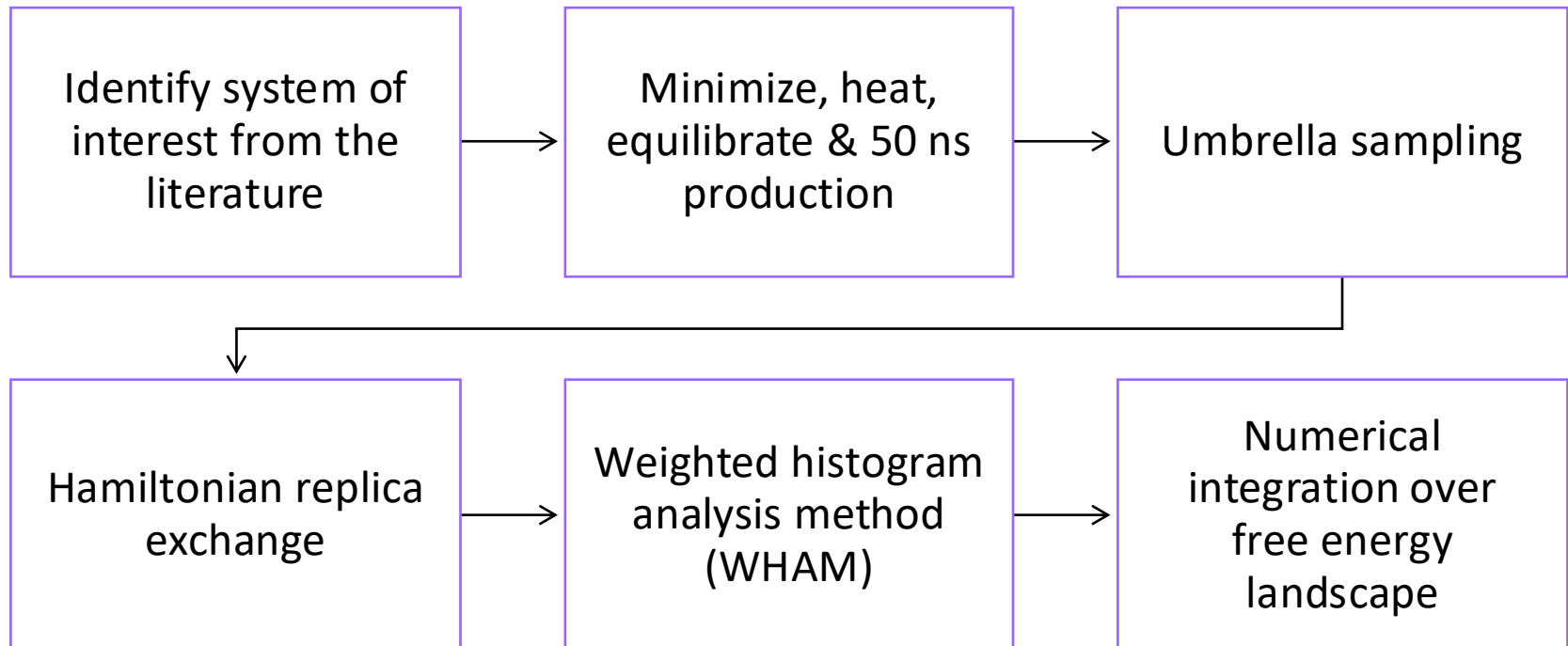
- Regulation
- Communication
- Structure
- ...

Allosteric regulations are crucial in biological processes

- Many PPIs (e.g., enzyme-substrate binding) require cofactors
- Free energy changes inform allosteric conformational shifts

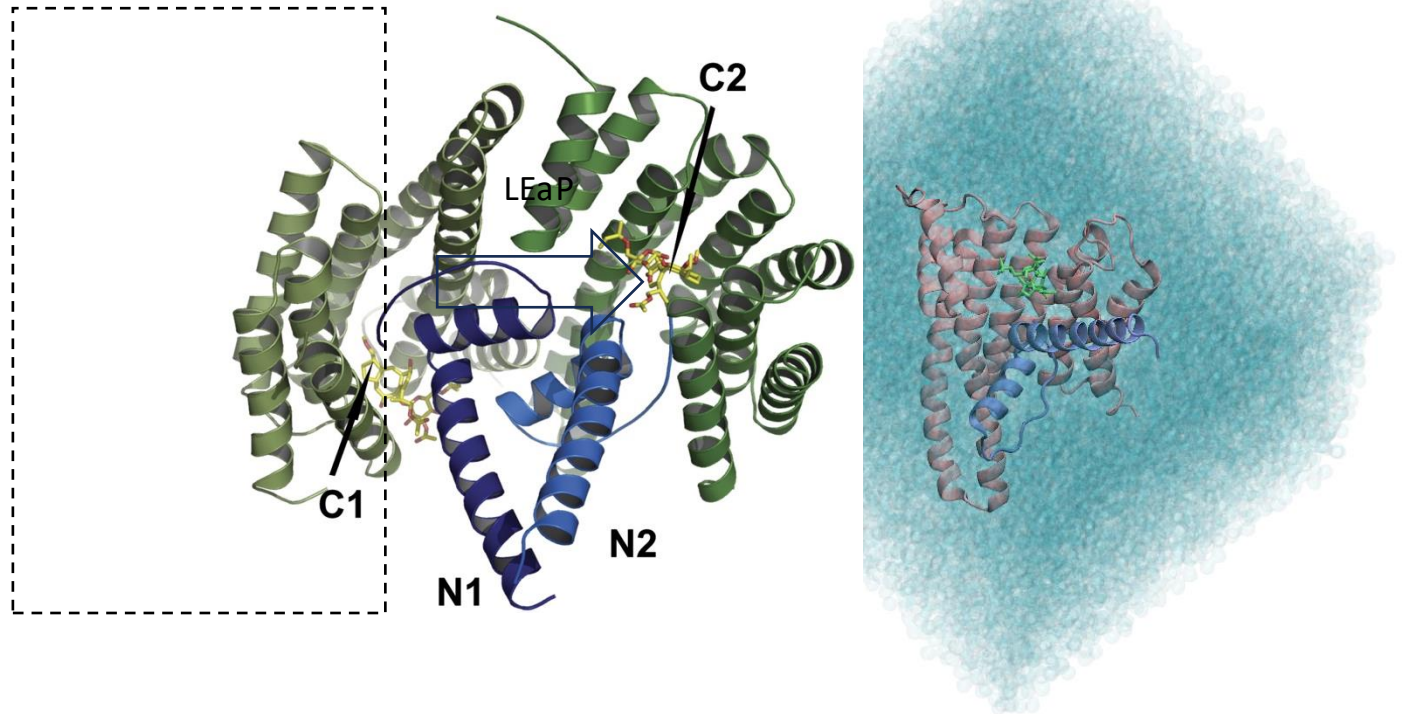


Workflow of this study



Model system

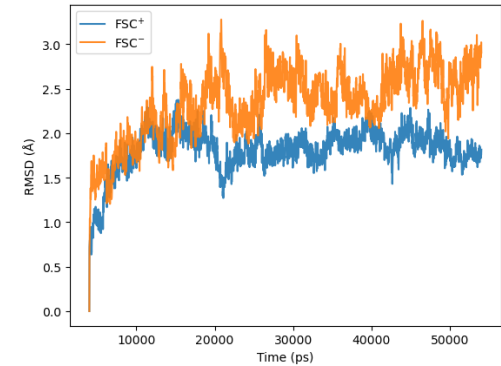
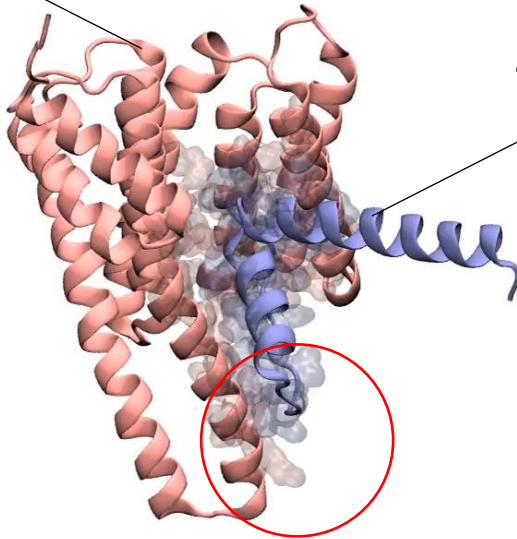
- 14-3-3 protein + plant plasma membrane H⁺-ATPase (N1, N2) + fusicoccin (C1, C2)
- Monomeric partners dissolved in cubic solvent box



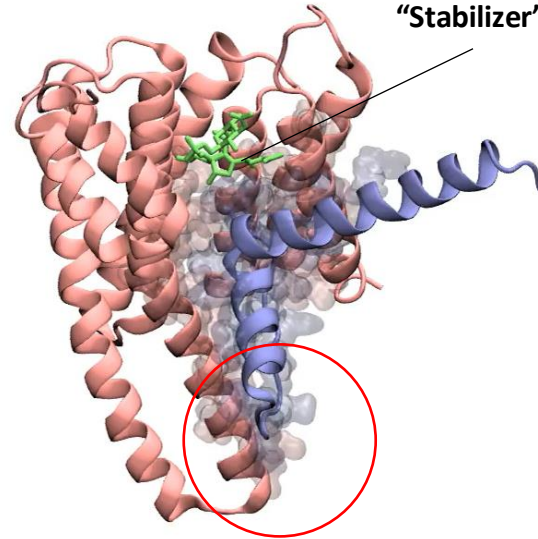
Fusicoccin increases complex interface rigidity

“Receptor” (R)

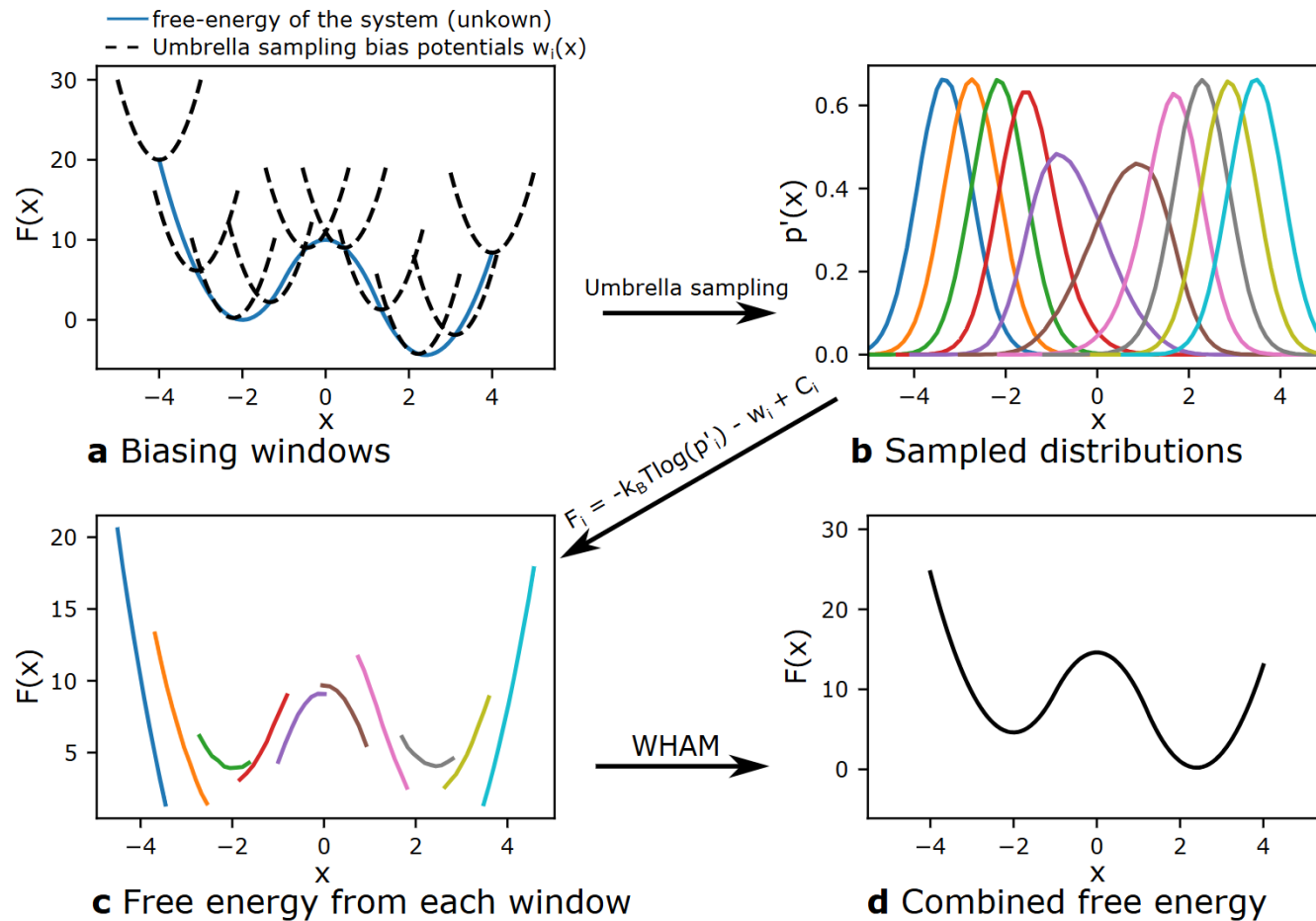
“Ligand” (L)



“Stabilizer” (S)

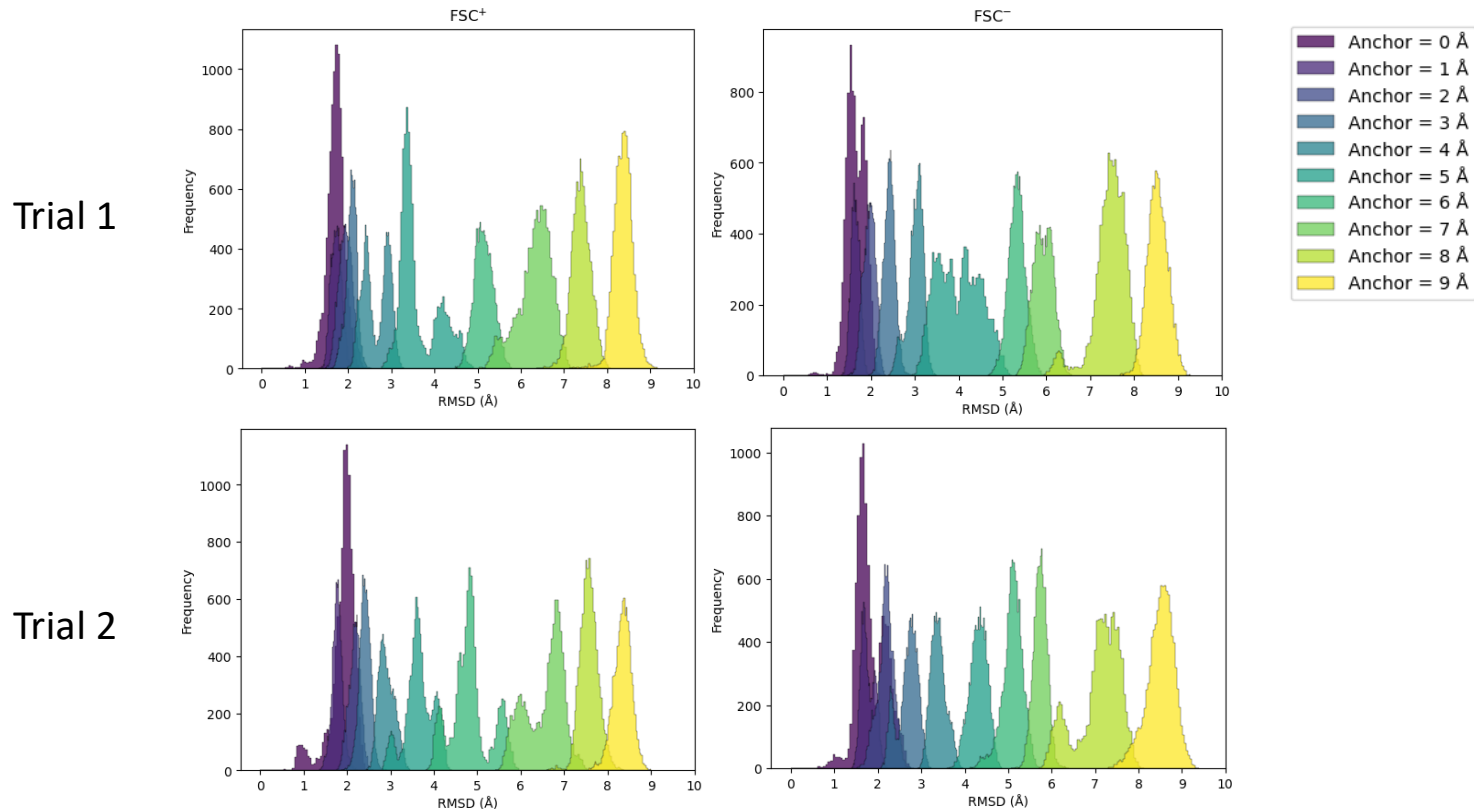


Umbrella sampling & WHAM



Umbrella sampling samples most conformations

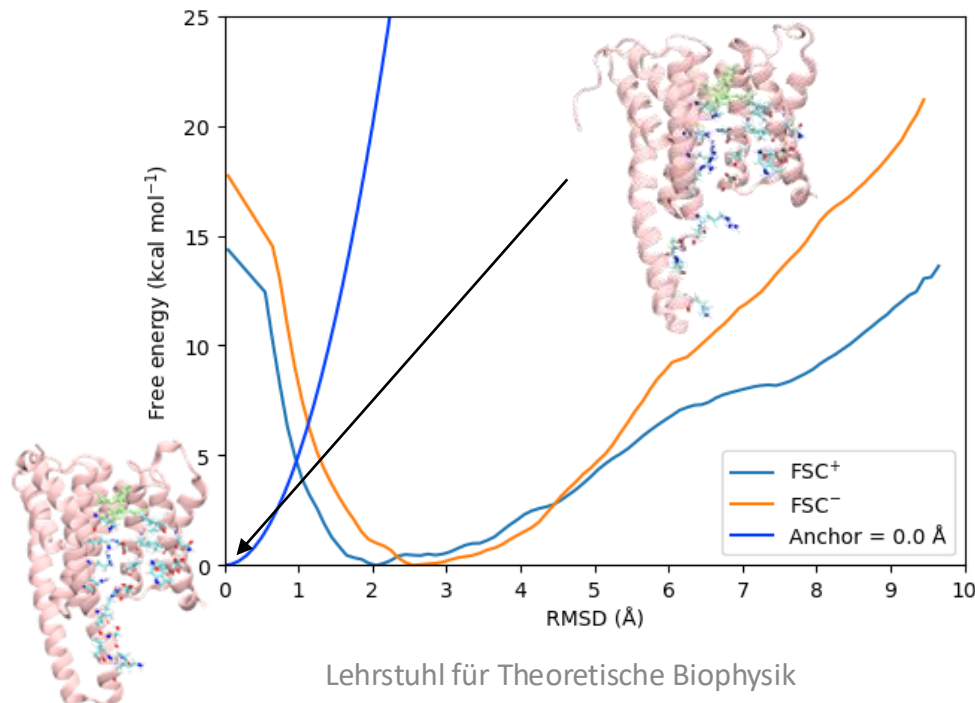
- Restraint on receptor and ligand (complex) interface atoms



Numerical integration of free energy landscape

- Free energy required to revert to the reference structure

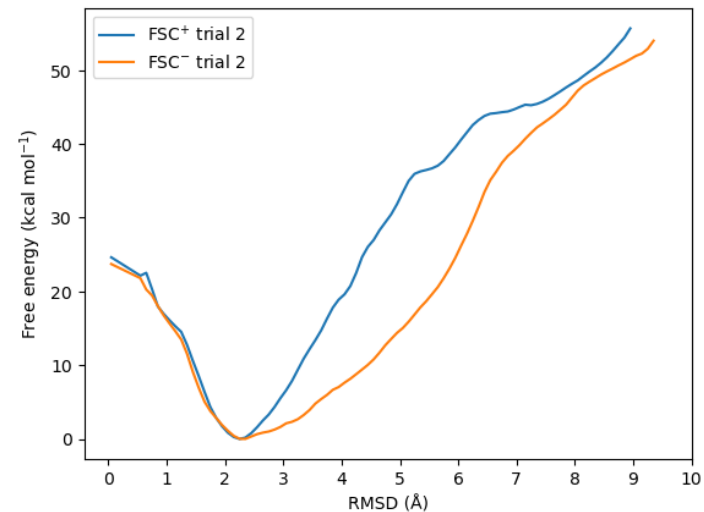
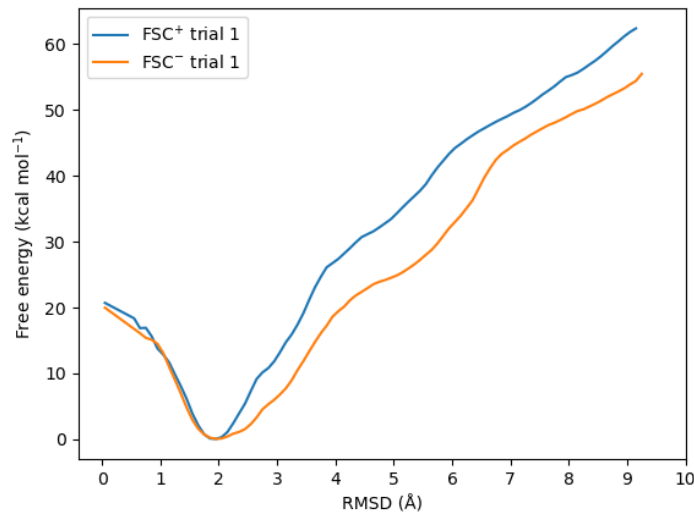
$$\Delta G = -RT \ln \frac{\int_{\min}^{\max} e^{-\frac{\beta}{N_A} G_{\text{RMSD}}} e^{-\frac{\beta k}{N_A} (\text{RMSD} - \text{RMSD}_{\text{ref}})^2} d(\text{RMSD})}{\int_{\min}^{\max} e^{-\frac{\beta}{N_A} G_{\text{RMSD}}} d(\text{RMSD})}$$



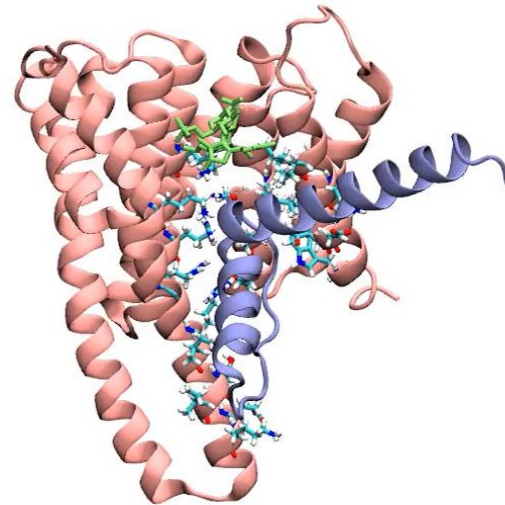
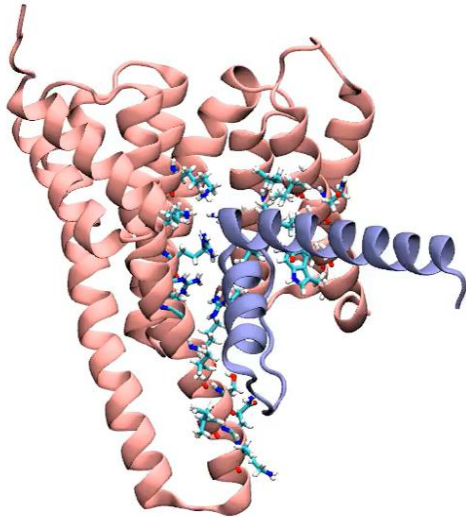
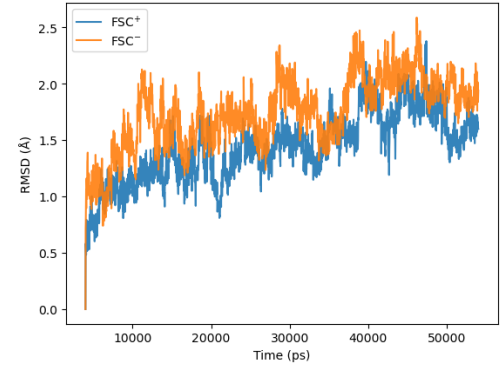
Complex interface do not show stabilizer-dependence

System	$\Delta G / \text{kcal mol}^{-1}$		
	Trial 1	Trial 2	
FSC^+	16 ± 1	20.2 ± 0.3	} no difference?
FSC^-	15 ± 1	19 ± 1	

Additional simulation for convergence?



Decoupling receptor and ligand protein interfaces



Restraining only receptor interface shows no dependence

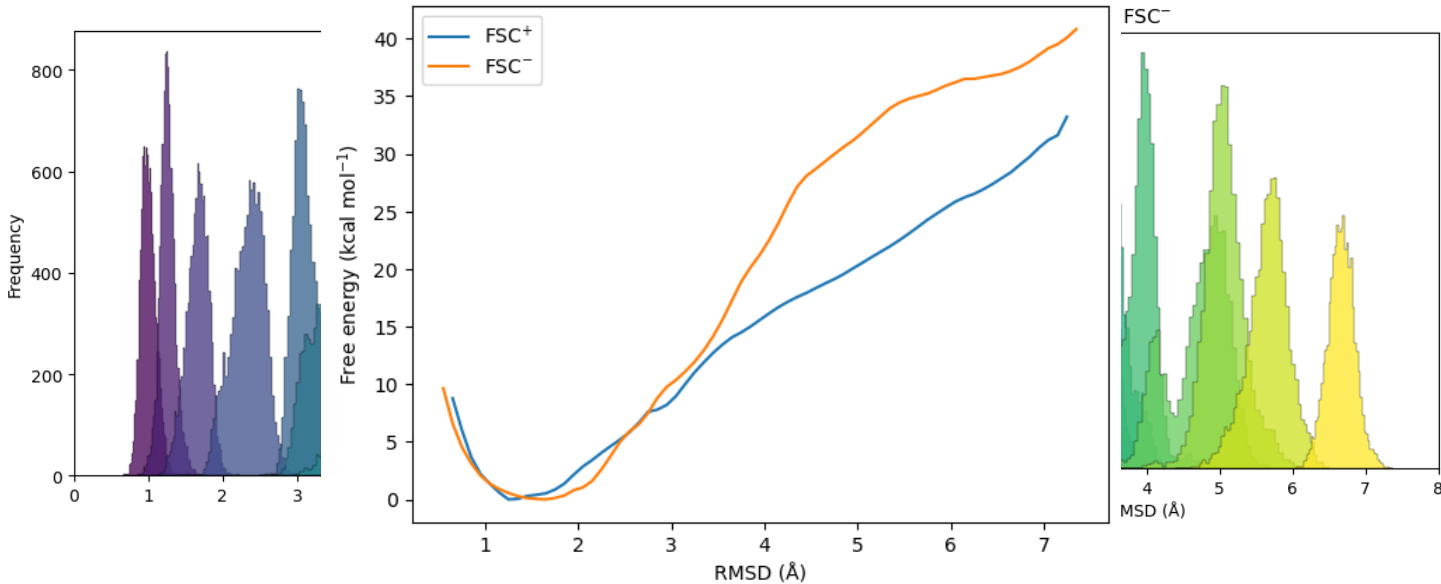
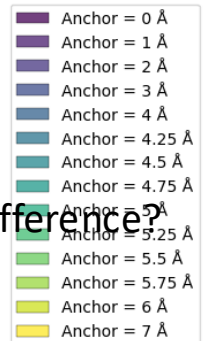
- More intermediate anchors (windows)

- Hamiltonian replica exchange umbrella sampling

System

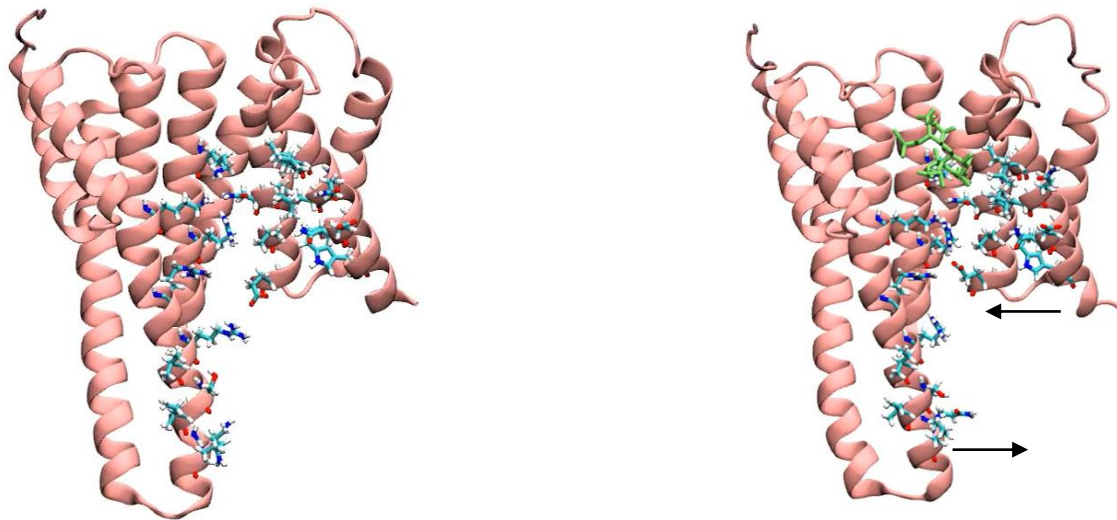
 $\Delta G / \text{kcal mol}^{-1}$ FSC⁺ 7.0010 ± 0.0006 FSC⁻ 7.0354 ± 0.0003

Still no difference?



H⁺-ATPase: a secondary stabilizer?

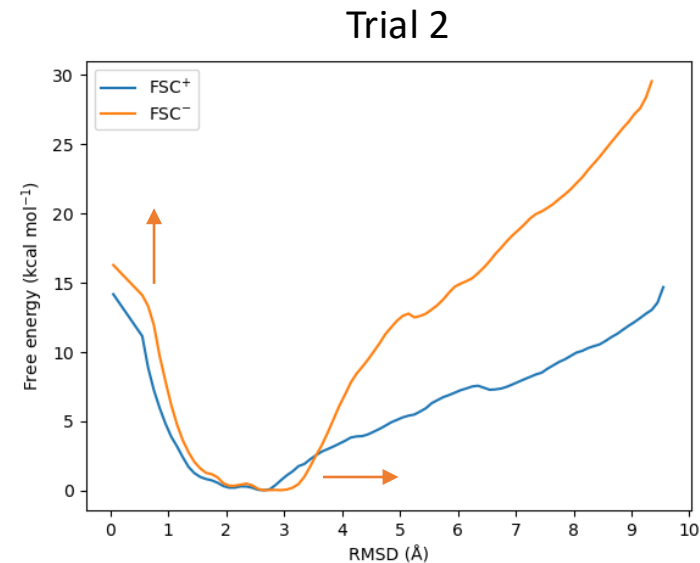
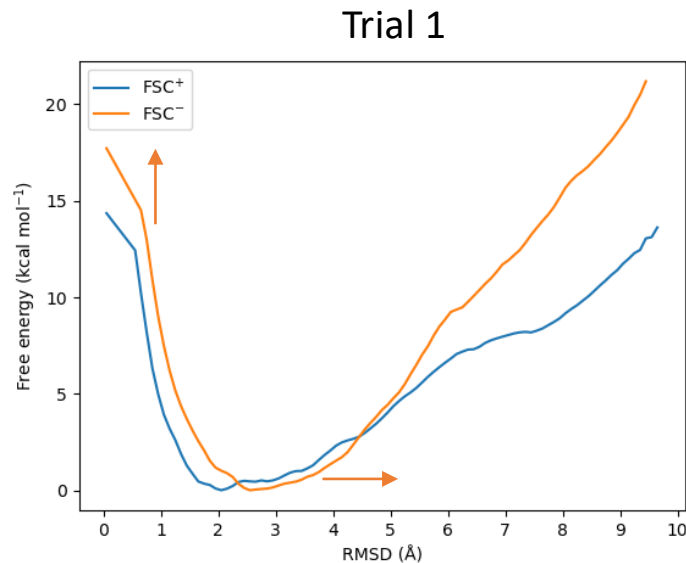
- Fusicoccin “preps” the 14-3-3 to bind to H⁺-ATPase



Removing H⁺-ATPase increases stabilization effect of fusicooccin

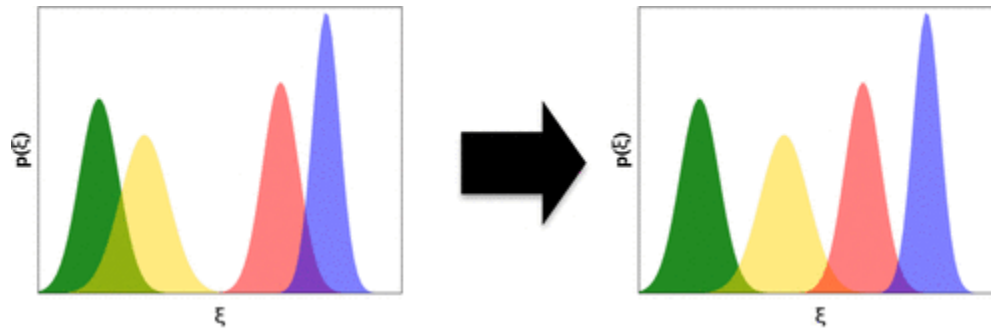
System	$\Delta G / \text{kcal mol}^{-1}$	
	Trial 1	Trial 2
FSC ⁺	10.18 ± 0.01	9.92 ± 0.02
FSC ⁻	14 ± 1	12 ± 1

} 2-4 kcal mol⁻¹



Future directions

- Validate fusicoccin's stabilization free energy on 14-3-3 protein
- Identify and simulate “better” systems (e.g., reported allostery)
- Improve overlap of umbrella sampling windows (esp. replica exchange)



- Replicates for statistical significance

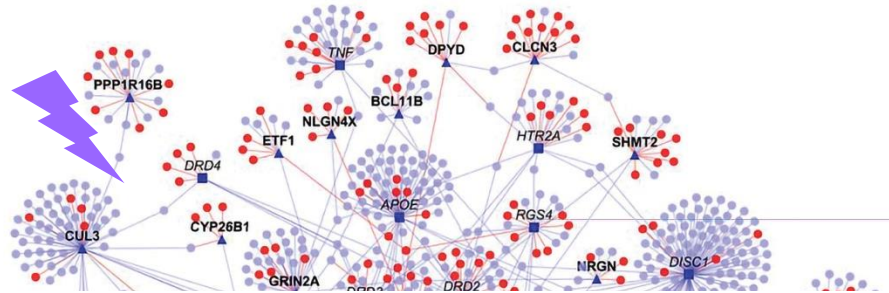
Thank you for listening & a great internship!

Specials thanks to:
Prof. Dr. Martin Zacharias
Dr. Ibrahim Mohd
Christian Sustay, MSc
Tristan Mauck
for helping me learn MD from scratch!

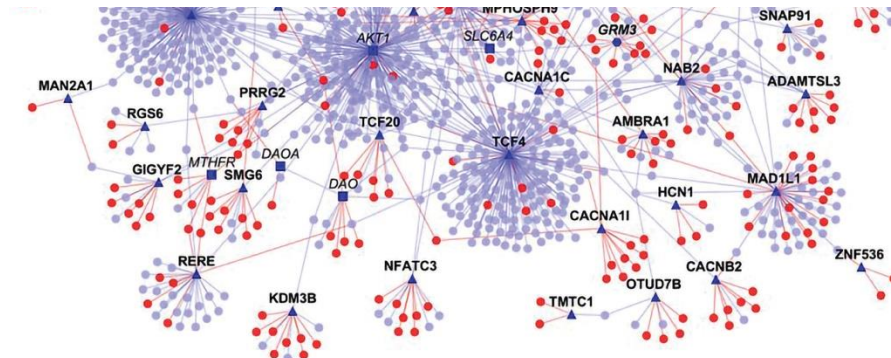
Appendix

Protein-protein interactions are biological regulators

- Disease: disruption to an existing interaction

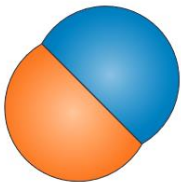
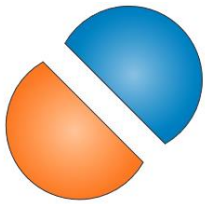


**Can molecules “repair”
disrupted interactions?**

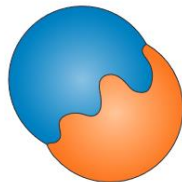
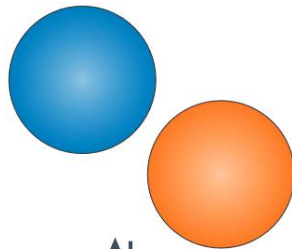


PPIs take many forms

two globular proteins with
preformed surfaces



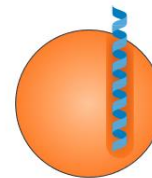
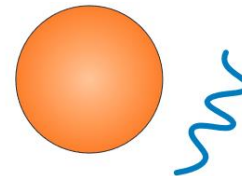
two globular proteins with an
induced binding surface



rigid globular protein
with a peptide



flexible globular protein
with a peptide

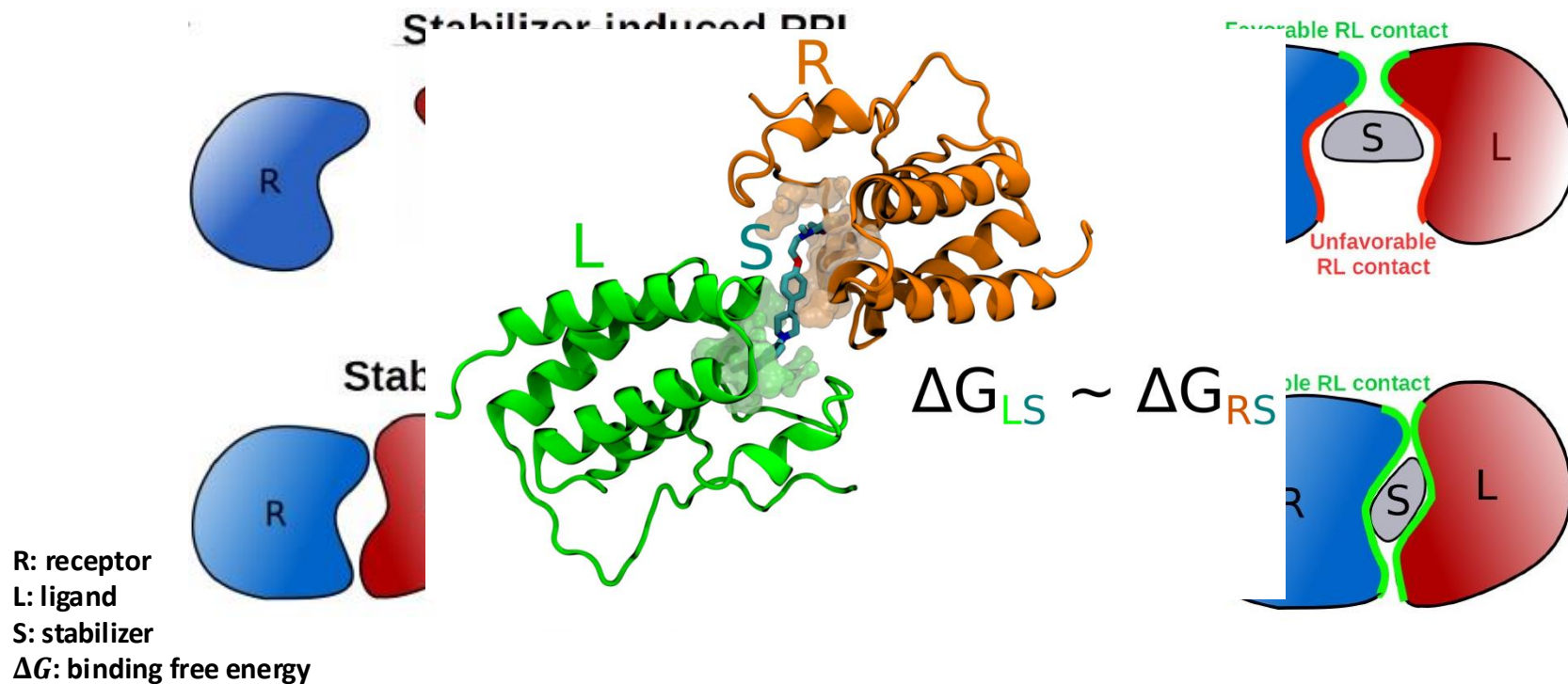


two peptides



PPIs often involve “stabilizers”

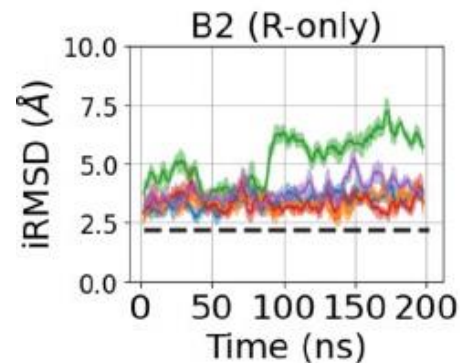
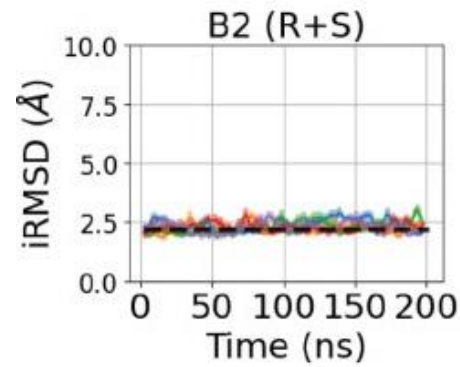
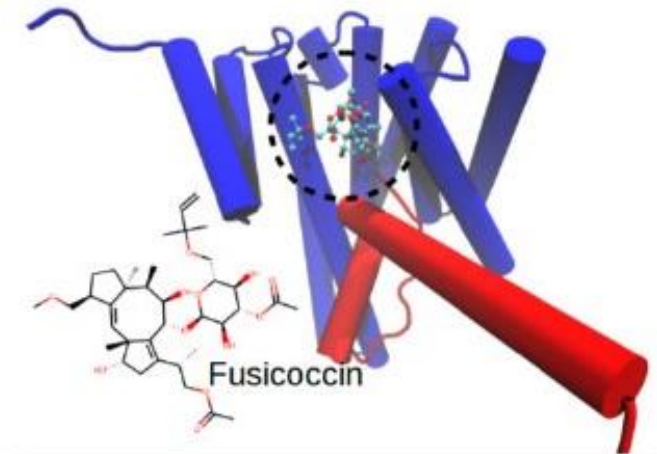
- “Dual-binding mechanism”
- Both protein partners bind to a smaller compound with similar affinity



Identifying model systems

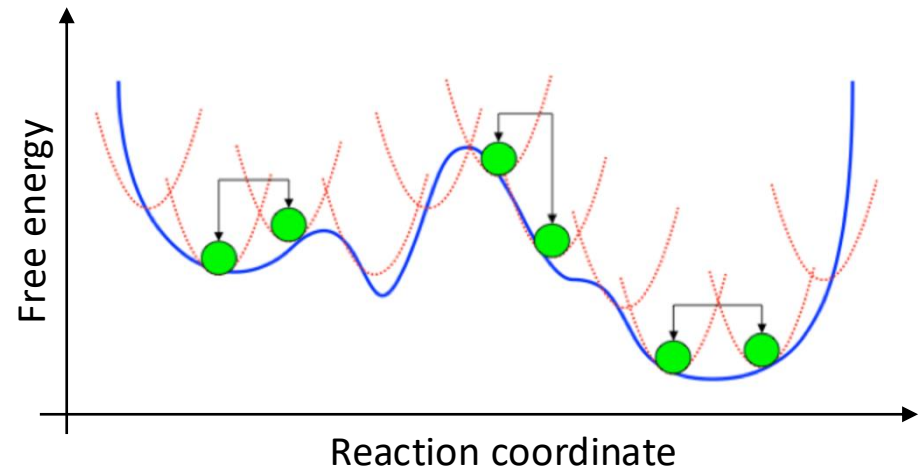
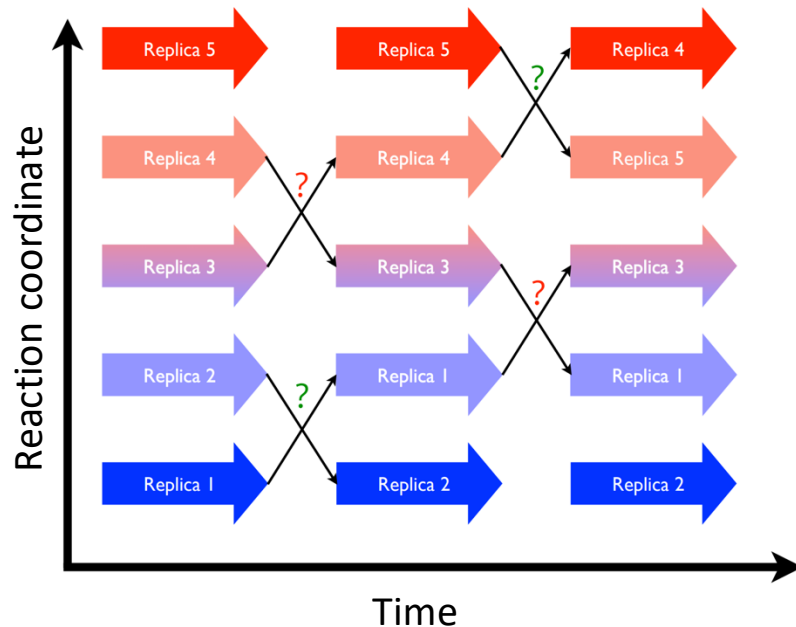
- Interface root-mean square deviation (iRMSD) fluctuates significantly without stabilizer

B2. 14-3-3A/H⁺-ATPase



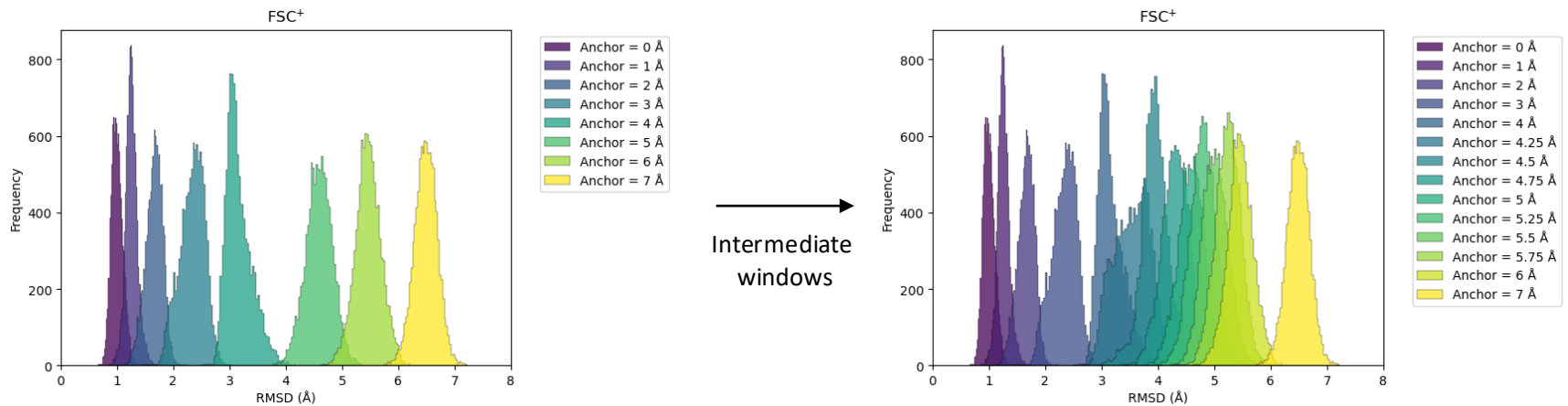
Hamiltonian replica exchange

- Builds on umbrella sampling
- Exchanges between neighbours with similar potential energy landscapes
 - Conformational space between windows sampled more efficiently

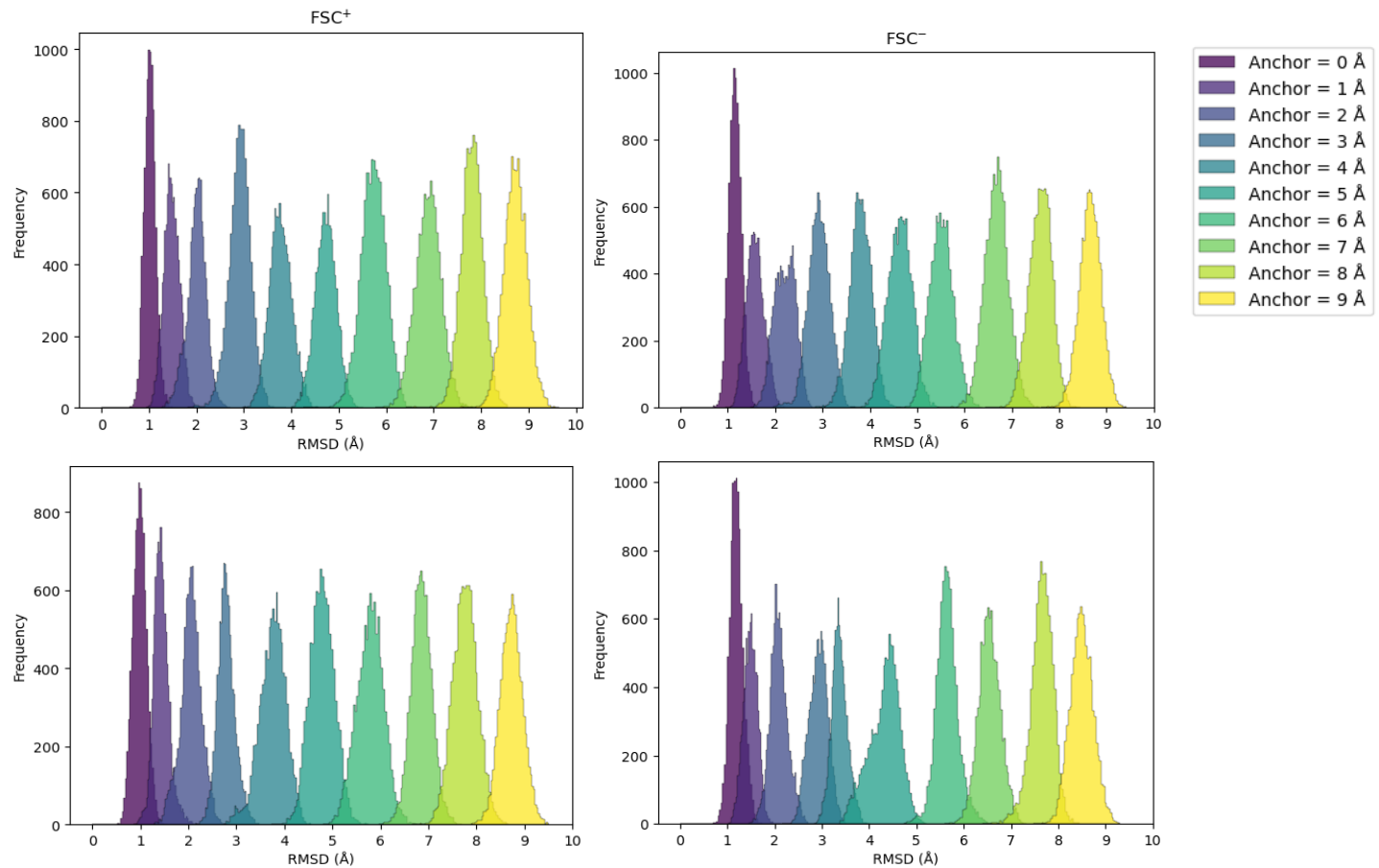


Increasing number of windows for better overlaps

- Sample rare events (e.g., local maxima)

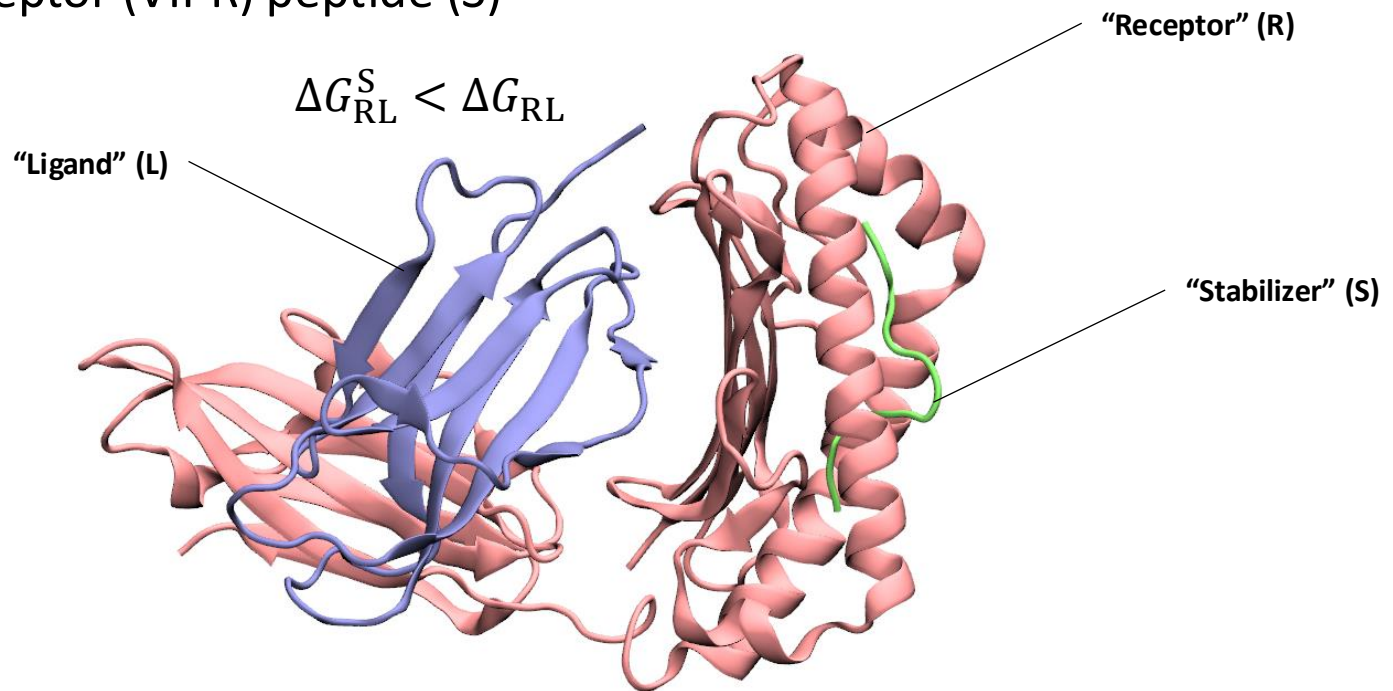


Sampling windows without H⁺-ATPase



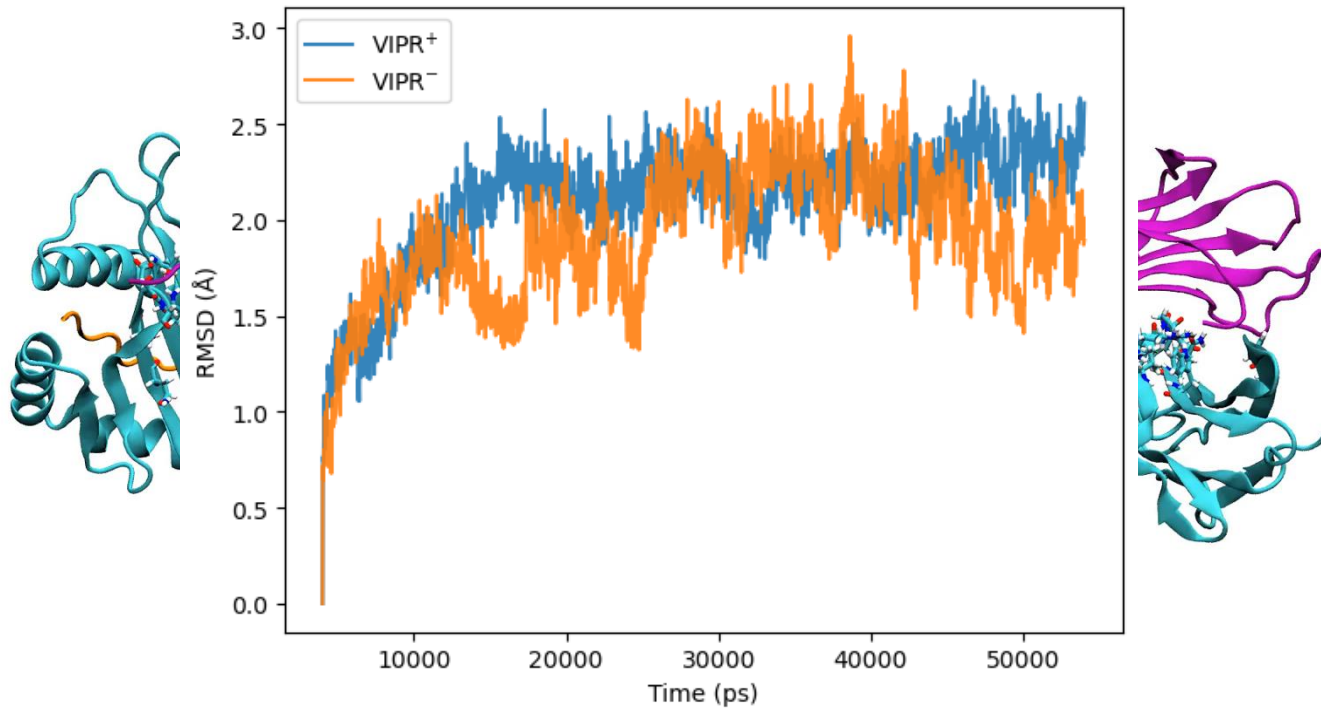
Investigating systems with “true” allosteric interactions

- Better systems: stabilizer only interacts with the receptor
- HLA (R) -B*2709 (L) complexed with the vasoactive intestinal peptide type 1 receptor (VIPR) peptide (S)

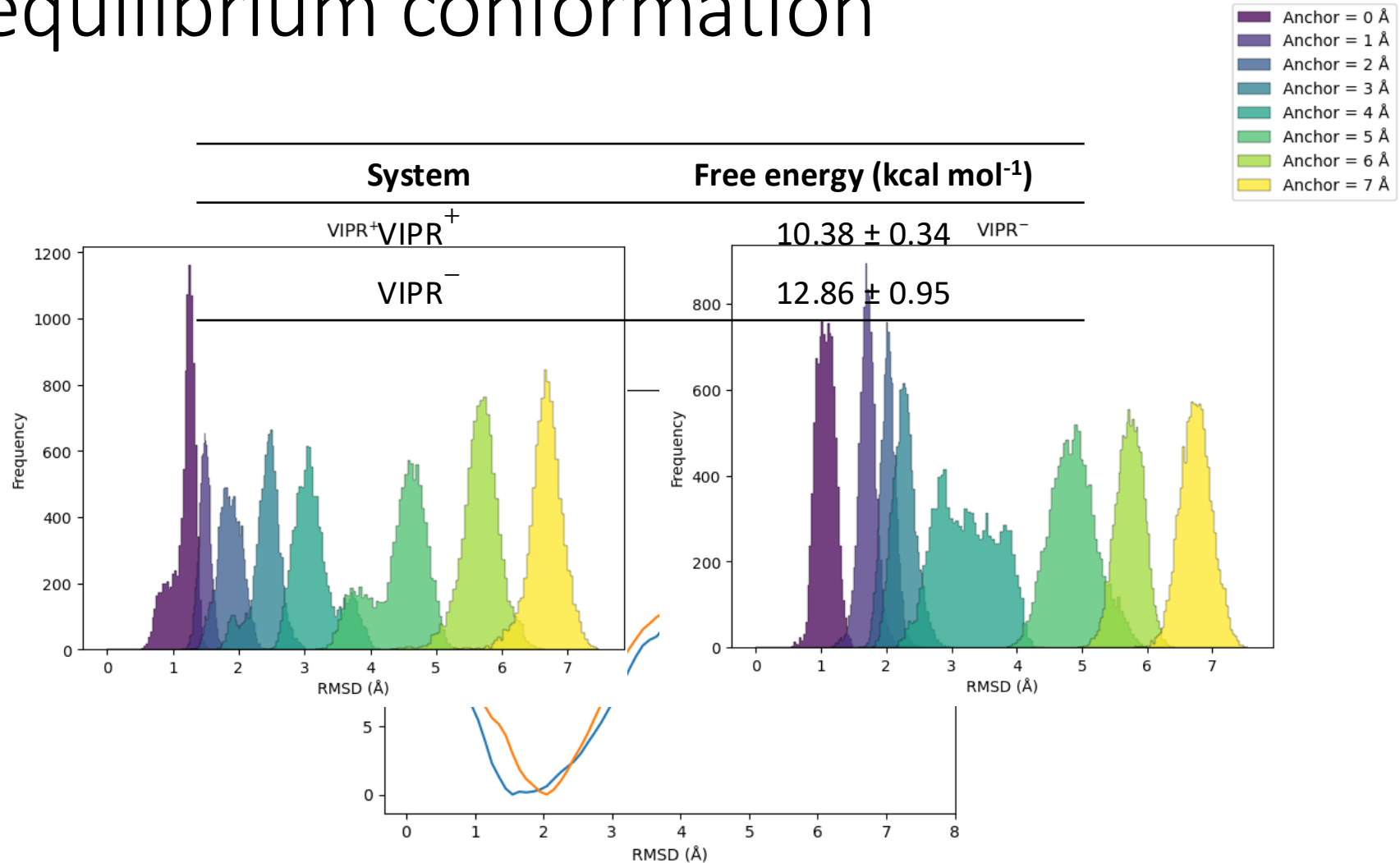


VIPR stabilizes HLA interface

- 50 ns NPT production MD

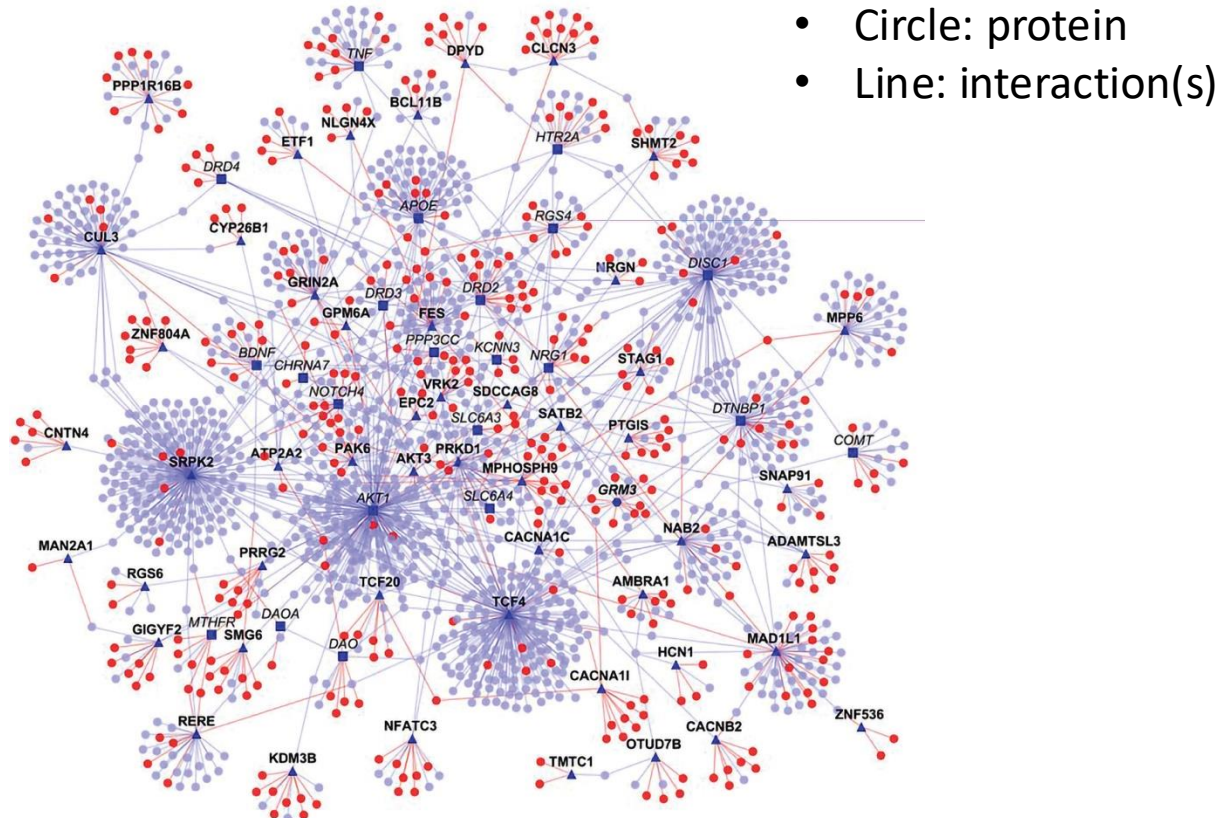


VIPR stabilizes HLA interface to equilibrium conformation



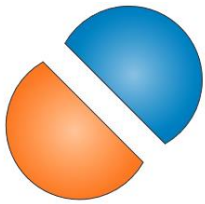
Past materials

Protein-protein interactions are biological regulators



PPIs take many forms

two globular proteins with
preformed surfaces



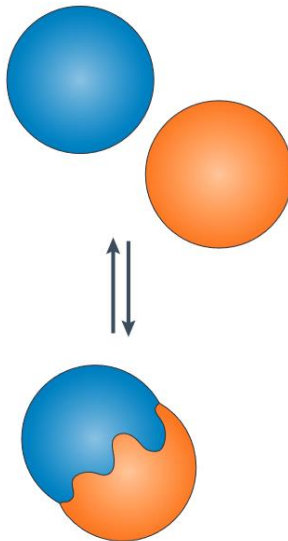
rigid globular protein
with a peptide



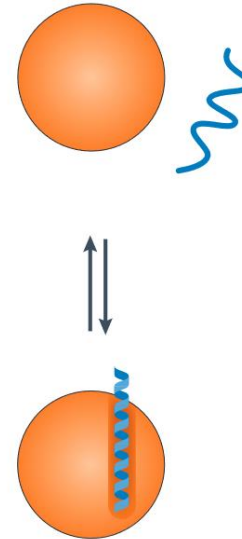
two peptides



two globular proteins with an
induced binding surface

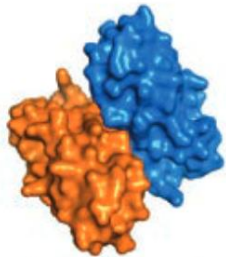


flexible globular protein
with a peptide

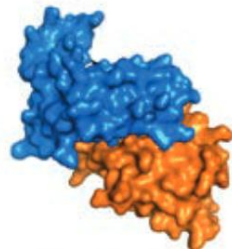


PPI examples

two globular proteins with
preformed surfaces



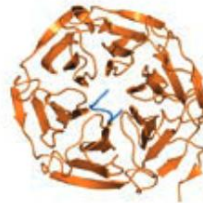
Cytochrome c'



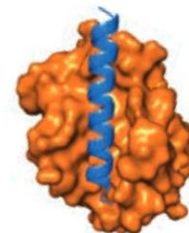
IL-2-IL-2R

two globular proteins with an
induced binding surface

rigid globular protein
with a peptide



KEAP1-NRF2



BCL-X_L-BAD

flexible globular protein
with a peptide

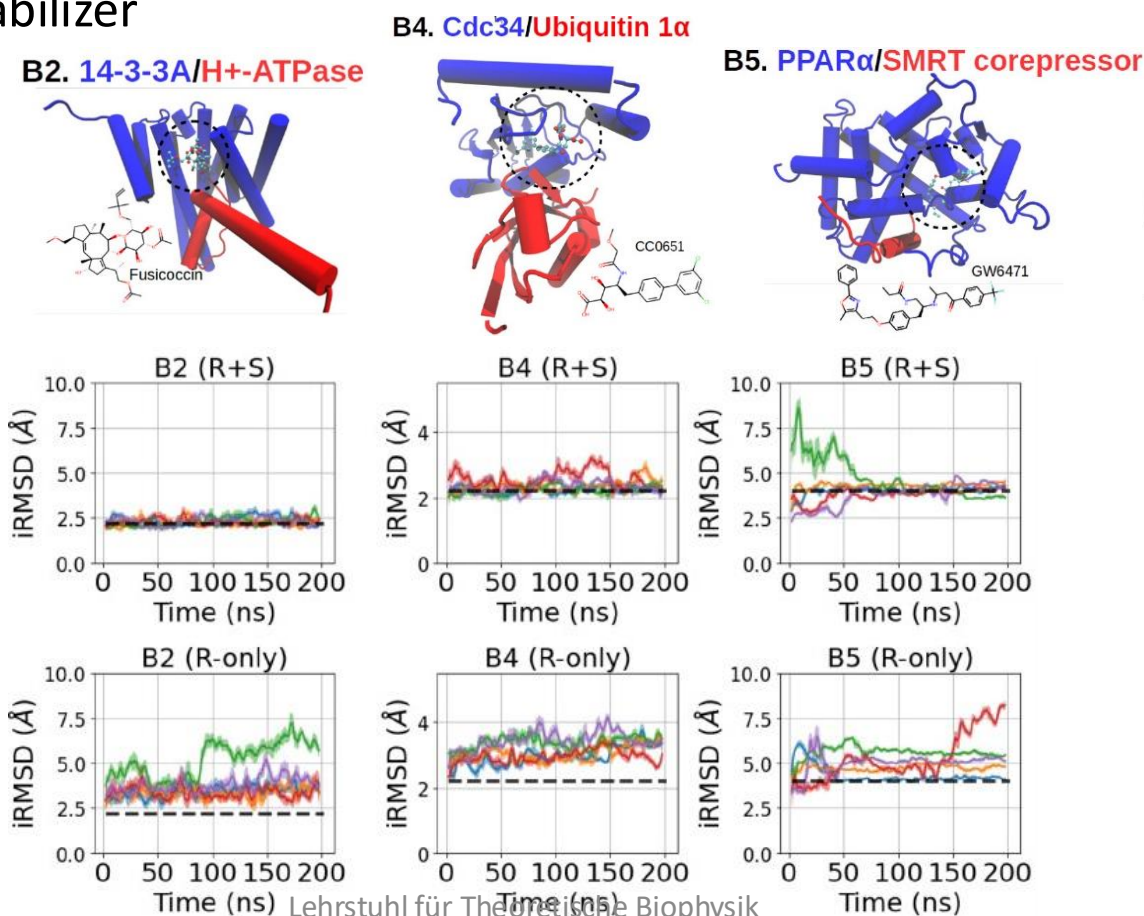
two peptides



MYC-MAX

Identifying model systems

- Interface root-mean square deviation (iRMSD) fluctuates significantly without stabilizer



Elucidating allosteric conformational changes

- Vary positional restraints on PPI interface

